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Author reply

Dear Editor:
With regard to Dr. Lee's question of a possible alternative diagnosis of arteritis in patients with enhancing magnetic resonance imaging lesions whom we presumed to have nonarteritic anterior ischemic optic neuropathy (NAION), we were mindful to exclude cases that were even suggestive of arteritis. Our description of methods was intended to make clear our attempt to identify and eliminate such cases.

We also were curious about the two cases of "NAION" that had enhancement of the optic nerve, which prompted us to review these cases again. Of the two patients with enhancement, neither reported symptoms consistent with giant cell arteritis or other causes of inflammatory or infectious optic neuropathy. The erythrocyte sedimentation rates were 7 and 22, respectively, and complete blood counts were normal. Angiotensin-converting enzyme, antinuclear antibody, and fluorescent treponemal antibody blood test results were normal in one patient (none of these were obtained in the other). Both patients had small cup/disc ratios. Follow-up examination did not uncover evidence of diagnoses more likely than NAION.

Interestingly, the magnetic resonance imaging scan of patient 56 was interpreted as normal by both the senior author and a neuroradiologist (other than the study neuroradiologist). Our study neuroradiologist (JDR) interpreted the scan as showing a small area of "slight enhancement" of the clinically affected optic nerve, which according to the design of the study was recorded as a positive scan. By coincidence, this scan was randomly selected for a re-review (by the same neuroradiologist) to determine the degree of consistency of interpretations, and the second interpretation was the same (i.e., slight enhancement). The variance among reviewers indicates that this abnormality was subtle and perhaps of questionable clinical significance.

We do not have an explanation for the apparent enhancement of the optic nerve on magnetic resonance imaging scans of patients who are presumed to have NAION. We accept the possibility that the radiologic findings might indicate an alternative diagnosis, although none has become evident in these two patients.

JOSEPH F. RIZZO, III, MD
CHRISTOPHER M. ANDREOLI, MD
JAMES D. RABINOV, MD
Boston, Massachusetts

Retinal Hemorrhages in Children

Dear Editor:
In Schloff et al's article,[1] all cases of suspected nonaccidental trauma were excluded. Because two of the authors in a previous article stated the finding of retinal hemorrhages in a child with a head injury suggests a nonaccidental cause,[2] the conclusion that retinal hemorrhage is uncommon in children with intracranial hemorrhage not resulting from shaken baby syndrome is unwarranted. The only trauma cases left in the study should have been those without retinal hemorrhage. Indeed, the 7-year-old should probably have been excluded, although the earlier article only studied children younger than 3. The correct conclusion is that very few children with retinal hemorrhage cannot be suspected as having been abused.

HORACE B. GARDNER, MD, JD
Manitou Springs, Colorado

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Author reply

Dear Editor:
Although we welcome Dr. Gardner's letter, we found it difficult to interpret his comments. We think he is concerned that the children in our study were preselected in such a way as to eliminate those children with retinal hemorrhages who might have been assumed to have been victims of the shaken baby syndrome.

We did exclude all cases of suspected nonaccidental trauma. However, this was done by the admitting hospital service *before* the child was examined by the ophthalmology team. Therefore, retinal hemorrhages were not used to predetermine the diagnosis. Interestingly, those children who were excluded from our study based on an initial suspicion of possible abuse (all children had their eye examinations within 72 hours of admission) but whose cases were later adjudicated by our child abuse team as accidental injuries had no retinal hemorrhages. Had they been included in our study, this would have further underscored the rarity of retinal hemorrhage associated with intracranial hemorrhage in children who are not victims of shaken baby syndrome.

Dr. Gardner refers to another publication from our center.[1] Although the intent of that study was not to specifically address the issue of retinal findings in children with intracranial hemorrhage, of the 79 children with head injury, 8 had intracranial hemorrhage. Three of those eight cases were nonaccidental trauma, and all had retinal hemorrhages. The remaining five cases of accidental trauma all showed no evidence of retinal hemorrhages, again supporting our conclusion that "retinal hemorrhage is uncommon in children with intracranial hemorrhage" not due to child abuse.

We agree that the diagnosis of shaken baby syndrome

must not rest solely on the presence of "retinal hemorrhage" in either the clinical or the research setting.[2] In addition to describing the type, location, distribution pattern, and number of retinal hemorrhages, one must also take into consideration the history (or absence of history) and the nature of other injuries, particularly to the bones and brain.

The medical problem of child abuse is discussed in a unique medicolegal context, so much so that some physicians have virtually made their careers out of testifying as expert witnesses. Legal and medical definitions of science and fact can be disturbingly different. Clarity of definition and adherence to proper medical scientific principles of investigation, including peer-review publication, remain an essential element in exploring and understanding the pathophysiology and potential forensic implications of findings such as retinal hemorrhage.

ALEX V. LEVIN, MD, FRCSC
SUSAN SCHLOFF, MD
PAUL B. MULLANEY, MD
DEREK C. ARMSTRONG, MD
EMMANUEL SIMANTIRAKIS, MDCM
ROBIN P. HUMPHREYS, MD FRCSC
JOHN S. MYSEROS, MD
J. RAYMOND BUNCIC, MD, FRCSC
Toronto, Canada

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Macular Hole Surgery

Dear Editor:

We greatly appreciate the reply to our letter by Dr. Da Mata and colleagues[1] and fully agree that more comparative data on functional outcome of indocyanine green (ICG)–assisted macular hole surgery are necessary. Therefore, our clinical and histopathologic results have been submitted for publication.

From a clinicopathologic point of view, one comment by Dr. Da Mata and colleagues needs further clarification. They used the term "surface of retinal ILM," which might be misleading, as the interpretation of histologic findings requires further information on which side of the internal limiting membrane (ILM) the authors refer to, the retinal or the vitreal side. When evaluating the ultrastructure of the ILM removed during vitrectomy, the key questions are the following:

1. Are cellular elements (such as inner portions of Müller cells) present on the retinal surface of the ILM?
2. If so, can this be attributed to ILM peeling itself or to the additional use of ICG?

In their reply Dr. Da Mata and colleagues mention the presence of cellular elements "on the surface of retinal ILM" in eyes with "coexisting epiretinal membranes." In these cases, it should not be surprising to find cellular elements on the vitreal side of the ILM, as shown by previous reports as well as by Dr. Da Mata and coworkers themselves.[2]

Regarding the retinal side of the ILM, there were no cellular elements present after ICG-assisted peeling, as shown in Figure 2 of the latter report.[2] In our experience, the absence of cellular elements on the retinal side of the ILM is consistent with ultrastructural analysis of specimens after ILM peeling, but *without* ICG assistance. In contrast, after ICG-assisted ILM removal, cellular elements of the inner retina adherent to the ILM were common features of histologic specimens.[3] In a previous study these findings were also observed in cadaver eyes but were considered a postmortem artifact.[4] The presence of cellular structures adherent to the ILM was further reported by Kwok and colleagues after ICG-guided vitrectomy.[5] However, from the figures provided in their report and due to the histologic technique applied, it was hardly possible to determine the real location of these cells.[6]

In summary, we emphasize that retinal structures had been found on the retinal surface of the ILM after ICG-assisted vitrectomy not only in cadaver eyes but also after ICG-guided macular hole surgery in humans. Attention should be paid to ultrastructural analysis of ILM specimens after ICG-assisted peeling, as this might further clarify recent controversial discussions on possible negative effects of ICG on the human retina.

CHRISTOS HARITOGLOU, MD
ARND GANDORFER, MD
ANSELM KAMPIK, MD
München, Germany

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Postoperative Wound Dehiscence

Dear Editor:

The article by Abou-Jaoude et al[1] brings to mind a postoperative complication after penetrating keratoplasty (PKP) that is rarely investigated. In this retrospective, noncomparative study of 69 eyes that underwent removal of a single running suture after PKP, the authors identified 5 that de-